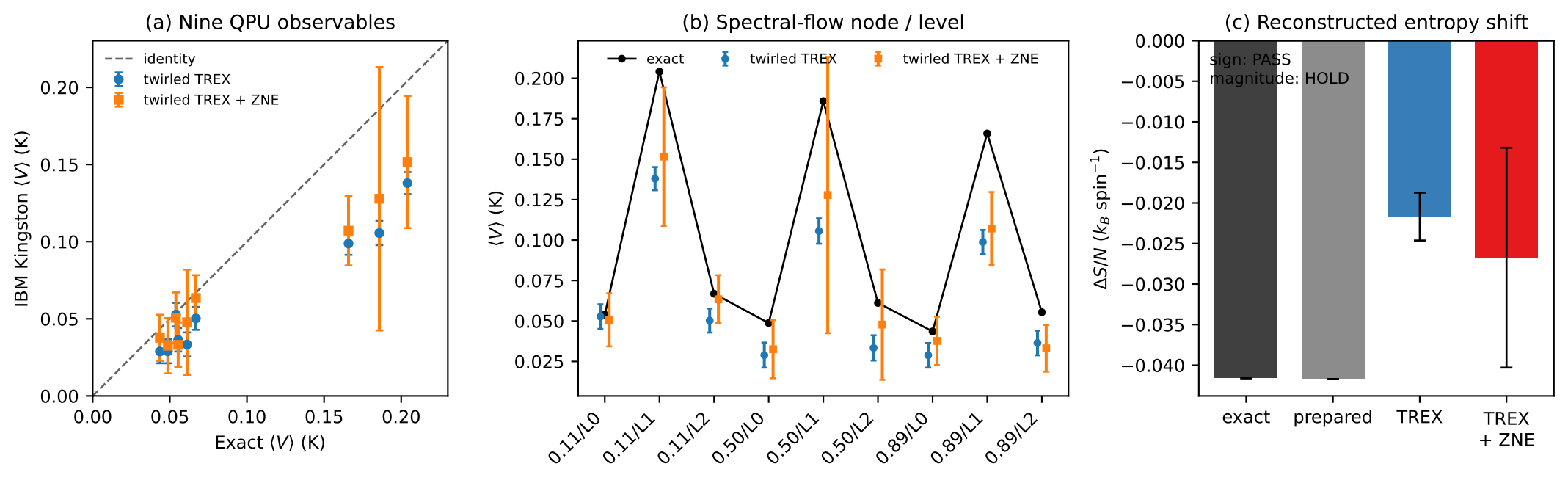
(a) Nine QPU observables
identity
twirled TREX
twirled TREX + ZNE
IBM Kingston ⟨V⟩ (K)
Exact ⟨V⟩ (K)
0.00
0.05
0.10
0.15
0.20
(b) Spectral-flow node / level
exact
twirled TREX
twirled TREX + ZNE
⟨V⟩ (K)
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0.11/L0
0.11/L1
0.11/L2
0.50/L0
0.50/L1
0.50/L2
0.89/L0
0.89/L1
0.89/L2
(c) Reconstructed entropy shift
sign: PASS
magnitude: HOLD
ΔS/N (k_B spin^−1)
−0.000
−0.005
−0.010
−0.015
−0.020
−0.025
−0.030
−0.035
−0.040
exact
prepared
TREX
TREX + ZNE